

Sepsis & Septic Shock

Early recognition and resuscitation for IM residents

Step 1 — Recognize organ dysfunction

Infection + dysfunction = sepsis; not just fever

- Sepsis = infection + life-threatening organ dysfunction (dysregulated host response)
- Septic shock = vasopressors to keep MAP ≥ 65 AND lactate > 2 despite fluids
- Dysfunction clues: hypotension, $\uparrow$ lactate, AKI/oliguria, altered mentation
- Also: $\uparrow$ bilirubin, thrombocytopenia, new oxygen requirement
- qSOFA is NOT a rule-out screen — low sensitivity

PEARL A normal qSOFA does not exclude sepsis — keep working it up.

Step 2 — The first hour

Culture, antibiotic, fluid — then reassess

Action	Detail
Blood cultures ×2	Before antibiotics — but never delay antibiotics to get them
Antibiotics	Broad-spectrum ≤1 h for shock; ~3 h window only if possible-not-probable & no shock
Fluids	~30 mL/kg balanced crystalloid (preferred over saline), then reassess
Lactate	Draw and trend as a perfusion marker

PEARL Source control beats any antibiotic — find and drain the focus on day one.

Step 3 — Pressors for the non-responder

Norepinephrine first; don't drown a fluid-unresponsive patient

Agent	Role
Norepinephrine	First-line; titrate to MAP ≥ 65 (peripheral OK short-term)
Vasopressin	Add-on; catecholamine-sparing
Epinephrine	Second add-on; consider with cardiac dysfunction
Hydrocortisone	Refractory shock on escalating pressors

PEARL Judge fluid need dynamically (passive leg raise, PPV) — not by CVP.

Source control & de-escalation

- Find the source: exam, imaging, cultures
- Control it early: drain abscess, remove infected line, relieve obstruction
- De-escalate antibiotics on culture data; stop if infection excluded
- Use procalcitonin trends to support stopping
- Many controlled sources need only ~5–7 days of therapy

PEARL Persistent shock on good antibiotics = look for an uncontrolled source.

Reassess — and what to avoid

DO

- Reassess MAP, lactate clearance, UOP, mentation
- Favor balanced crystalloid
- Start norepinephrine early for non-responders
- Plan source control + de-escalation from day 1

AVOID

- qSOFA to rule out sepsis
- Fixed fluid boluses if fluid-unresponsive
- Starch colloids (AKI/mortality)
- Routine antipyretics, beta-blockers, probiotics (2026 SSC)

PEARL Fluid is a drug with a dose — reassess after every bolus.

Take-home points

- Sepsis = infection + organ dysfunction; shock adds pressor need + lactate >2
- Hour 1: cultures (don't delay abx), early antibiotics, balanced fluids, lactate
- Dose fluids to dynamic responsiveness, not CVP
- Norepinephrine first-line; vasopressin/steroids for refractory shock
- Source control + de-escalation matter as much as the first dose